Lab Procedure
Ultrasonic

Introduction

Ensure the following:

1. You have reviewed the **Application Guide – Ultrasonic**
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
   a. The LED next to the USB C port will stay white after the touch screen starts loading.
   b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.
3. Have a tape measure that can measure a distance of at least 1 m, as well as some different objects for testing the ultrasonic sensor, with different shapes and materials, including at least:
   a. A large, hard, flat object, such as a book, tablet, or a laptop.
   b. A cylindrical object, such as a disposable or reusable water bottle, that is less wide than the flat object.

Exploring Ultrasonic Sensing

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File > Open Folder**, and browse to the directory containing the python files and lab procedure for the Ultrasonic lab (…/sensors_trainer/1_fundamentals /ultrasonic /hardware/python). Open this directory.
2. Open **ultrasonic_single_distance.py**.
3. We will explore using the ultrasonic sensor to measure the distance from the Sensors Trainer to different objects. Position the Sensors Trainer so it is standing upright. Set up a tape measure to measure the distance from the sensor.
4. Click ▷ to run the code. This will open a scope displaying the measured distance from the ultrasonic sensor. Here we will test the minimum distance that the sensor can reliably read. Take the flat object and slowly move it away from (and towards) the sensor while keeping it parallel to the Sensors Trainer to a distance up to 70cm while observing the scope. Using the tape measure, what is the minimum distance from the sensor that the object can accurately detected? Observe the terminal, where the

measured distances are printed as outputs. How consistent is the measured distance reading when you leave the device and the object in the same place?

5. Stop the script by clicking back on the terminal in Visual Studio Code, then pressing **Ctrl + C**, and then clicking enter when prompted in the terminal.

6. Now we will test the maximum distance that the sensor can reliably read. Run the code and move the flat object farther from the sensor while observing the scope. What happens to the distance signal as the flat object is moved farther?

7. The ultrasonic sensor in the Mechatronics Sensors Trainer is the ULTRA-CH201-00ABR. Open the datasheet for this sensor located in the ultrasonic sensor directory (…/sensors_trainer/1_fundamentals /ultrasonic) and report the operating range listed on the first page. Does this match with your observed minimum and maximum ranges?

8. Run the code again. Observe the scope as you move the flat object towards and away from the sensor at different speeds. Try orienting the object at different angles to the Sensors Trainer or placing the object at an offset so it is not directly in front of the sensor. Describe your observations. Stop the code.

9. In the code, find the **`Experiment constants`** region, change **`ultraDistanceMax`** from 5 to 0.5. Now when **`SensorsTrainer`** is initialized with **`ultraLength = ultraDistanceMax – ultraDistanceMin`**, this sets **`ultraLength`** to 0.5. Run the code and move the flat object between 0 and 0.7 m from the sensor while observing the scope. How did the readings change when you decreased the ultrasonic sensor range? What situations would call for a shorter detection range or a farther one? Stop the code.

10. Open ultrasonic_IQ.py. This code runs the ultrasonic sensor on I/Q mode and generates 2 plots: Ultrasonic IQ, and Ultrasonic Amplitude. In this section, we will focus on the Ultrasonic Amplitude scope, which plots the signal strength (amplitude) by distance from the sensor. You will explore placing different objects at varying distances from the sensor to see how it affects the amplitude at different distances. In this code, the device is initialized with **`ultraService`** set to 2 to run the ultrasonic sensor on IQ service mode rather than distance mode (**`ultraService = 1`**).

    *Note: In-Phase and Quadrature (I/Q or IQ) data is a method of modulating sinusoidal waves that allows the sensor to capture distance data with more precision. Refer to **Concept Review – Radar** to learn more about IQ data components and how to obtain useful information from IQ data.*

11. Click ▷ to run the code. Place the water bottle 0.4m from the sensor. How does this appear on the amplitude plot?

12. Try placing objects with varying colors, shapes, sizes, and materials in front of the ultrasonic sensor at different distances while observing the shape of the amplitude plot. For example, compare the plots for a water bottle and a flat object placed at the same distance from the sensor. Describe how the different properties of the object affect the shape of the plot. How does the signal appear for the same object at different distances? Include some screenshots to show your observations.

13. Scroll to the section of the datasheet on Typical Operating Characteristics that shows how the signal amplitude varies across the field-of-view. (Note: this plot is for a different module than the one on the Sensors Trainer but both modules have a 45-degree field-of-view). Consider the note under the figure in the parentheses. Scroll to another section on Object Detection. How do these two sections explain your observations in this lab? Discuss any factors that influenced the operating range.

14. Stop, save and close your program.

15. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.